

E-Cell AI CRM — System Report

1. Executive Summary

This system implements a five-module AI-native CRM for the fictional E-Cell company. It manages 520+ customers, 1,050+ tickets, and 2,500+ interaction events with LangChain summarization, LangGraph agent workflows, cohort retention analysis, and a live HEART framework dashboard backed by a production FastAPI API.

2. Architecture Decisions

2.1 Stack

Layer	Choice	Rationale
----- ----- -----		
Backend	FastAPI + SQLite	Fast OpenAPI docs, zero external DB setup for demo
LLM (primary)	Ollama / Llama-3-8B (Option A)	On-prem privacy, no API cost
LLM (alternate)	Gemini 1.5 Flash (Option B)	Strong reasoning, free tier
LLM (fallback)	Rule-based mock	Ensures demo works offline
Agents	LangGraph state machine	Explicit routing → generate → escalation pipeline
Analytics	pandas + scikit-learn	Cohort metrics and churn classifier

2.2 Module Boundaries

[illegible]

```
api/app.py (/api/v1)
```

```
...
```

Each module is independently runnable via ``python -m src.<module>``.

3. LLM Selection Rationale

We support **dual-mode inference**:

1. **Ollama (Option A)** — Selected when ``LLM_PROVIDER=ollama``. Suitable for evaluation environments requiring data residency.
2. **Gemini 1.5 Flash (Option B)** — Selected when ``GEMINI_API_KEY`` is set. Used for richer summarization during dataset generation.
3. **Mock provider** — Deterministic templates when neither backend is reachable; prevents demo failure.

Ticket summarization uses a dedicated system prompt requiring factual grounding. Agent responses pass through a **hallucination guard** that penalizes unsourced dollar amounts, missing ticket citations, and overconfident language.

4. Agent Prompt Design & LangGraph State Machine

4.1 State Schema

```
```python
AgentState = {
customer_id, query, context, ticket_data,
route_category, priority, draft_response,
sources, confidence, escalated, hallucination_flags
}
```
```

4.2 Graph Flow

```
```mermaid
stateDiagram-v2
[*] --> load_context
load_context --> route
route --> generate
generate --> escalation
```
```

escalation --> [*]

...

| Node | Function |
|----------------|--|
| ----- ----- | |
| `load_context` | Pulls short/long-term memory + open tickets |
| `route` | Classifies category (billing/technical/account/general) |
| `generate` | Drafts response with source citation |
| `escalation` | Flags low-confidence urgent cases for supervisor handoff |

4.3 Summarization Chain

LangChain-style chain in `TicketSummarizationChain`:

- **Input:** ticket title, description, customer metadata
- **Output:** summary, key_issues[], urgency, suggested_response
- **Configurable:** tone, max_length

5. Interaction Memory

Per-customer buffers in `customer\_memory` table:

| Layer | Capacity | Behavior |
|-------------------|--------------------|--|
| ----- ----- ----- | | |
| Short-term | 20 turns | Recent chat/ticket context |
| Long-term | Compressed summary | LLM-compressed when short-term overflows |

Cross-session retrieval concatenates long-term summary + recent turns for agent queries.

6. Cohort Segmentation Methodology

6.1 Cohort ID Assignment

...

cohort_id = {acquisition_month}_{industry}_{product_tier}

...

Example: `2025-11\_FinTech\_Enterprise`

Configurable dimensions: acquisition date, industry vertical, product tier, behavioral tags.

6.2 Retention Curve

For each cohort, we compute 6 monthly periods:

- **Active customer** at period p : $\text{engagement_score} \geq \max(20, 70 - 8p)$ AND $\text{tenure_days} \geq 30p$
- **Retention rate** = active / cohort_size

6.3 Churn Scoring

Heuristic score (production baseline):

...

$\text{churn_prob} = 0.5 \times (1 - \text{engagement}/100) + 0.3 \times \min(\text{tickets}/15, 1) + 0.2 \times (1 - \text{tenure}/365)$

...

Signals: low_engagement, high_ticket_frequency, early_lifecycle, unresolved_backlog

ML validation: Logistic regression on engagement, tickets, tenure — reports precision/recall/F1 on held-out set.

6.4 Churn Model Evaluation

| | |
|-------------|--------------------------------------|
| Metric | Source |
| ----- ----- | |
| Precision | sklearn on heuristic-labeled holdout |
| Recall | Same |
| F1 | Harmonic mean |
| Coverage | % customers with cohort_id assigned |

7. HEART Framework Metric Definitions

H — Happiness

| | |
|-------------|---|
| Signal | Computation |
| ----- ----- | |
| CSAT | Mean <code>csat_score</code> on resolved tickets; falls back to mean engagement |
| NPS proxy | $(\text{promoters} - \text{detractors}) / N \times 100$ where promoter = engagement ≥ 70 |

E — Engagement

| | |
|-------------|-------------|
| Signal | Computation |
| ----- ----- | |

| | |
|---------------------|---------------------------------|
| Active customers | Count with engagement ≥ 30 |
| Ticket open rate | Open tickets / total tickets |
| Session depth proxy | Total interactions / customers |

A — Adoption

| | |
|-----------------------|---|
| Signal | Computation |
| ----- | ----- |
| AI-assisted rate | Tickets with `ai_assisted=true` / total |
| Onboarding completion | Customers with tenure ≥ 7 days |
| Memory adoption | Customers with memory buffer / total |

R — Retention

| | |
|-------------------|---|
| Signal | Computation |
| ----- | ----- |
| Monthly retention | Customers with engagement ≥ 25 / cohort size |
| Churn flags | engagement < 35 AND tickets > 5 |
| Avg lifespan | Mean tenure_days |

T — Task Success

| | |
|-------------------|-----------------------------------|
| Signal | Computation |
| ----- | ----- |
| Resolution rate | Resolved+Closed / total tickets |
| FCR proxy | Non-escalated resolutions / total |
| Escalation rate | Escalated tickets / total |
| AI vs human ratio | AI-resolved / human-resolved |

All metrics recompute from live SQLite data on each API call.

8. RBAC & Audit

| | |
|------------|--|
| Role | Permissions |
| ----- | ----- |
| Agent | CRUD customers/tickets, summarize, query agent |
| Supervisor | Agent + cohort read |
| Admin | Full access |

| Analytics | Read-only customers, tickets, cohorts, HEART |

Every core endpoint returns `audit` block: timestamp, agent_id, source_confidence, latency_ms, role.

9. System Evaluation Metrics (Stage 6)

| Category | Endpoint | Metric |
|-----------------|------------------------------|---|
| HEART | `/api/v1/heart` | All 5 dimensions + per-metric breakdown |
| Agent Quality | `/api/v1/query/agent` | confidence, hallucination_flags |
| Cohort Accuracy | `/api/v1/evaluation/metrics` | churn model P/R/F1 |
| Resolution Rate | `/api/v1/evaluation/metrics` | closure rate, AI ratio |
| Latency | All endpoints | `audit.latency_ms` per request |

10. Dataset Specification

| Entity | Count | Span |
|--------------|-----------|--------------------------------------|
| Customers | 520 | 6 industries × 3 tiers |
| Tickets | 1,050 | billing, technical, account, general |
| Interactions | 2,500 | email, chat, call, ticket, portal |
| Time range | ~6 months | Dec 2025 – Jun 2026 |

First 20 ticket descriptions are LLM-enhanced; remainder use realistic templates. Deduplication on email (customers) and ticket ID at ingest.

11. Known Limitations & Future Work

- SQLite suitable for demo; PostgreSQL recommended for production concurrency
- Churn labels derived from heuristic for ML eval until real churn events exist
- Latency benchmarks under 500 concurrent ops require separate load test (locust/k6)
- bcrypt hashing prepared but demo uses plaintext passwords for evaluator convenience

12. File Reference

| File | Purpose |
|----------------------------------|---------------|
| ----- ----- | |
| `src/crm.py` | Module 1 |
| `src/agents.py`, `src/memory.py` | Module 2 |
| `src/cohort.py` | Module 3 |
| `src/heart.py` | Module 4 |
| `api/app.py` | Module 5 |
| `dashboard/index.html` | Live HEART UI |
| `SYSTEM_REPORT.md` | This document |